AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M-MT131
Discrete Mathematics

Midterm Examination (MTA)
Spring Semester 2024/2025

Date: 8 /April/2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 90 Minutes

Instructions:

- Total Marks: **30**
- This exam consists of **four questions**.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

AOU EXAM PAPER

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Q–1: [5×3 marks] Choose the correct answer:

1.1 A propositional equivalent to $(p \wedge \neg q) \vee q \vee (\neg p \wedge q)$ is:

a) $\neg p \wedge q$
b) $p \wedge q$
c) $p \vee q$
d) $p \vee \neg q$
e) None of the mentioned

1.2 The statement," Every comedian is funny" where $C(x)$ is "x is a comedian" and $F(x)$ is "x is funny" and the domain consists of all people.

a) $\exists x(C(x) \wedge F(x))$
b) $\forall x(C(x) \wedge F(x))$
c) $\exists x(C(x) \rightarrow F(x))$
d) $\forall x(C(x) \rightarrow F(x))$
e) None of the mentioned

1.3 For two sets C and D the set $(C - D) \cap D$ will be _____

a) C
b) D
c) $C - D$
d) $\emptyset$
e) None of the mentioned

1.4 The value of $\lfloor \frac{3}{4} . \lfloor \frac{7}{3} \rfloor \rfloor$ is

a) 1
b) 2
c) 3
d) 0.75
e) None of the mentioned

1.5 The decimal expansion of $(2AE0B)_{16}$ is?

a) $(175627)_{10}$
b) $(175624)_{10}$
c) $(178566)_{10}$
d) $(175622)_{10}$
e) None of the mentioned

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Each question in below worth 15 marks. Answer all the following:

Q–2:

a) **[6 marks]** By constructing the truth table, Determine whether the following propositions is a tautology, contradiction or contingency: $[p \wedge (p \rightarrow q)] \rightarrow q$.

b) **[5 marks]** Without using truth tables show that : $(p \rightarrow q) \wedge (q \rightarrow \neg p)$ and $\neg p$ are logically equivalent .

c) **[4 marks]** Suppose that $P(x)$ is the statement "$2x - 1 = x$",where the domain of x is the set of integers. Determine the truth value of:

i. $\forall x \neg P(x)$

ii. $\exists x P(x)$

Q–3:

a) **[2+2+2+2 marks]** Assume that the universal set is the set of integers. Let $A = \{-3, -2, 2, 3\}$, $B = \{x \in Z \mid x^2 \leq 9\}$, $C = \{1, 2, 3, 4\}$. Find the following

i) $(A - B) \cup C$

ii) $(A \cap C)$

iii) Find $|P(A)|$

iv) Find $|A \times B|$

b) **[4 marks]** Write the set $A = \{1, 4, 9, 16, 25, \ldots\}$ in set-builder form.

c) **[2+1 marks]** Consider the following Arithmetic sequence: 50, 47, 44, 41,.....

i. Find the formula for the nth term of the sequence.

ii. find the first negative term.

Q–4:

a) **[5 marks]** Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$. Find $A \odot B$ and $B \odot A$.

b) **[3 marks]** Write the prime factorizations of 84 and 126, then find their GCD (Greatest Common Divisor) and LCM (Least Common Multiple).

c) **[7 marks]** A message has been encrypted using the function $(x)=(x+10)$ mod $26, 0 \leq x \leq 25$. Encode the message "NOT FAIR".

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	12	22	23	24	25

End of questions

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